06. Model Selection

Compare Random Forest, SVM, a neural network and imbalance-aware alternatives.

Project: Child Malnutrition Status Classification (ICFDR ML Project)

Source archive: ICFDR---ML-Project-main.zip | Main notebook: MalnutritionDetect.ipynb | Additional notebooks: SMOTE.ipynb and Oversampling on the whole dataset.ipynb

Random Forest

Random Forest trains many decision trees on bootstrap samples and random feature subsets, then aggregates their predictions. It handles nonlinear interactions and mixed-scale numeric inputs without requiring feature standardization. The notebook uses 300 trees in the main branch and 100 trees in the resampled-data notebook.

Support Vector Machine

A linear SVM finds a separating hyperplane with maximum margin; kernel SVMs use functions such as RBF to model nonlinear boundaries. The main notebook uses a linear SVC. The SMOTE notebook standardizes features and searches $C \in \{0.1,1,10\}$ and kernel$\in$ {linear, poly, rbf}; the saved best configuration shown in output is C=10, RBF.

Neural network branch

A small dense network with 64 and 32 ReLU units is included. However, it has a single sigmoid output and binary_crossentropy loss despite the target having four classes. This is a fundamental formulation mismatch. A four-class network should use four logits/softmax outputs with sparse_categorical_crossentropy or categorical_crossentropy.

Naive Bayes

A MultinomialNB experiment appears only as commented code. It is not appropriate for raw continuous measurements without a meaningful nonnegative count-like representation; GaussianNB would be a more natural simple probabilistic baseline for continuous features.

Recommended baseline comparison

Model	Strength	Caution
Majority baseline	Simple reference score	Misses all minority classes
Decision tree	Interpretable nonlinear splits	High variance on tiny dataset
Random Forest	Robust nonlinear ensemble	Can still favour majority class
SVM RBF	Flexible margin-based boundary	Needs scaling and careful tuning
GaussianNB	Fast probabilistic baseline	Strong distribution assumptions
Gradient boosting	Can model nonlinear interactions	Tune carefully; tiny data limits confidence

Selection principle

Choose based on macro-F1, per-class recall and confusion matrix, not accuracy alone. In a health-screening context, missing a rare class may be much more consequential than a false alert, but that cost must be defined with clinical/domain experts rather than assumed by the model developer.